

B.Tech. Degree III Semester Examination November 2013

SE 304 MANUFACTURING PROCESSES (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) How does steel differ from pig iron?
(b) Define the term “heat treatment” and state its objectives.
(c) What is resistance welding? What are its types?
(d) Differentiate between arc welding and gas welding.
(e) What are the various types of patterns?
(f) What is slush casting?
(g) Differentiate between cutting tool and machine tool.
(h) Sketch and explain cluster rolling mill.

PART B

(4 × 15 = 60)

- II. (a) How are engineering materials classified? Explain. (7)
(b) Write notes on: (8)
(i) Hardening
(ii) Normalising

OR

- III. (a) Explain Fe-C equilibrium diagram with neat sketch. (10)
(b) Write note on chemical properties of a material. (5)

- IV. (a) Sketch a gas welding set and label its parts. Show various types of flames with temperature. (7)
(b) Briefly explain various welding defects and inspection. (8)

OR

- V. (a) Compare TIG and MIG welding. (7)
(b) Briefly explain structural changes in HAZ when welding steel with neat sketch. (8)

- VI. (a) Explain the need for providing pattern allowance. (7)
(b) Explain the advantages and disadvantages of metal casting processes. (8)

OR

- VII. (a) Briefly explain casting defects and inspection. (7)
(b) Write notes on: (8)
(i) Die casting
(ii) Investment casting

- VIII. (a) Distinguish between hot working and cold working. (7)
(b) Explain orthogonal and oblique cutting with sketch. (8)

OR

- IX. (a) Briefly explain various methods of extrusion. (7)
(b) Write notes on ECM and EDM. (8)
